MNS Department
Semester: Summer-2025; Course ID: PHY 111
Course Title: Principles of Physics I; Total Marks: 15

Solution of Assignment-1

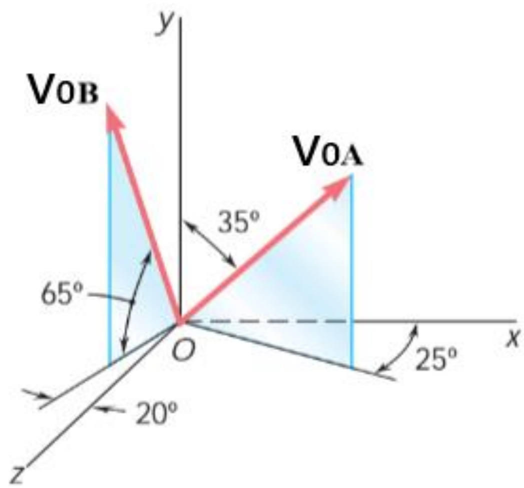

Fig. 1

1. Two particles A and B are projected from point O at the same time with the magnitude of initial velocities $\overrightarrow{V_{0A}} = 32\ m/s$ and $\overrightarrow{V_{0B}} = 40\ m/s$ as shown in Fig. 1. Consider vertical direction as Y axis and horizontal direction as XZ plane. Initial velocities and their components are make some angles with the Cartesian coordinate as shown in the Fig. 1. Both particles strike the ground in different location and embedded into the ground. Neglect the air resistance for the following calculations.

   (a) (2 marks) Calculate the time of flight of the particles.

   (b) (3 marks) Calculate the velocities of the particles at $t = 2\ sec.$

   (c) (3 marks) Find the final position of the particles with respect to point O and calculate the displacement of the particles after being striking the ground.

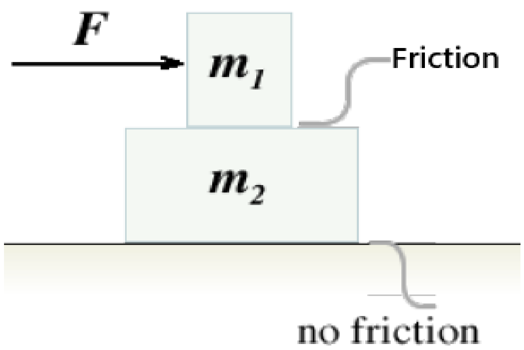

Fig. 2

2. Two blocks of masses $m_1 = 2.2\ kg$ and $m_2 = 4.8\ kg$ are stacked on top of each other and start at rest on the surface of a frictionless table as shown in Fig. 2. A force $\vec{F}$ is applied on block m_1 and coefficient of frictions between the blocks are $\mu_s = 0.4$ and $\mu_k = 0.3$.

   (a) (2 marks) Draw force diagrams for each of the blocks, clearly indicating all horizontal and vertical forces acting on them.

   (b) (3 marks) Find $\vec{F}_{max}$, the maximum value of $\vec{F}$ for which the upper block can be pushed horizontally so that the two blocks move together without slipping and find the acceleration of the system in this condition.

   (c) (2 marks) Assume that the magnitude of the force $\vec{F}$ is changed to $2\vec{F}_{max}$ and blocks do slip relative to each other. Determine each block's acceleration.

Solution of Assignment - 1

(1.) (a) **Particle-A**

$$y - y_0 = V_{oy} t + \tfrac{1}{2} a_y t^2$$

$$\Rightarrow 0 - 0 = 32\cos 35^\circ t - \tfrac{1}{2} \times 9.8 \times t^2$$

$$\Rightarrow t(26.213 - 4.9t) = 0$$

So, $t = 0$ and $26.213 - 4.9t = 0$

$$\Rightarrow t = \frac{26.213}{4.9} = \boxed{5.35\ sec}$$

$V_{oy} = V_{oA}\cos 35^\circ = 32\cos 35^\circ$

$a_y = -g = -9.8\ ms^{-2}$

Particle-B

$$y - y_0 = V_{oy} t + \tfrac{1}{2} a_y t^2$$

$$\Rightarrow 0 - 0 = 40\sin 65^\circ t - \tfrac{1}{2} \times 9.8 \times t^2$$

$$\Rightarrow t(36.25 - 4.9t) = 0$$

So, $t = 0$ and, $36.25 - 4.9t = 0$

$$\Rightarrow t = \frac{36.25}{4.9} = \boxed{7.4\ sec}$$

$V_{oy} = V_{oB}\sin 65^\circ = 40\sin 65^\circ$

$a_y = -g = -9.8\ ms^{-2}$

(b) **Particle-A**

At 2sec, $\vec{V_A} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$

$$\Rightarrow \vec{V_A} = V_{ox}\hat{i} + (V_{oy} + a_y t)\hat{j} + V_{oz}\hat{k}$$

$$\Rightarrow \vec{V_A} = 16.64\hat{i} + (26.213 - 9.8 \times 2)\hat{j} + 7.76\hat{k}$$

$$\Rightarrow \boxed{\vec{V_A} = 16.64\hat{i} + 6.613\hat{j} + 7.76\hat{k}}$$

$(V_0)_{xz} = V_{oA}\sin 35^\circ = 32\sin 35^\circ$

$V_{ox} = (V_0)_{xz}\cos 25^\circ = 32\sin 35^\circ\cos 25^\circ$
$= 16.64$

$V_{oz} = (V_0)_{xz}\sin 25^\circ = 32\sin 35^\circ\sin 25^\circ$
$= 7.76$

$V_{oy} = 32\cos 35^\circ = 26.213$

Particle-B

At 2sec, $\vec{V_B} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$

$$\Rightarrow \vec{V_B} = V_{ox}\hat{i} + (V_{oy} + a_y t)\hat{j} + V_{oz}\hat{k}$$

$$\Rightarrow \vec{V_B} = -5.78\hat{i} + (36.25 - 9.8 \times 2)\hat{j} + 15.89\hat{k}$$

$$\Rightarrow \boxed{\vec{V_B} = -5.78\hat{i} + 16.65\hat{j} + 15.89\hat{k}}$$

$(V_0)_{xz} = V_{oB}\cos 65^\circ = 40\cos 65^\circ = 16.91$

$V_{ox} = -(V_0)_{xz}\sin 20^\circ = -16.91\sin 20^\circ$
$= -5.78$

$V_{oz} = (V_0)_{xz}\cos 20^\circ = 16.91\cos 20^\circ$
$= 15.89$

$V_{oy} = 40\sin 65^\circ = 36.25$

(c) Particle-A

Position at 5.35 sec, $\vec{r}_A = x\hat{i} + y\hat{j} + z\hat{k}$

$x - x_0 = V_{0x} t$
$\Rightarrow x = 0 + (16.64 \times 5.35)$
$\therefore x = 89.024$

$y = 0$

$z - z_0 = V_{0z} t$
$\Rightarrow z = 0 + (7.76 \times 5.35)$
$\therefore z = 41.52$

$\therefore \vec{r}_A = 89.024\hat{i} + 41.52\hat{k}$

Particle-B

Position at 7.4 sec, $\vec{r}_B = x\hat{i} + y\hat{j} + z\hat{k}$

$x - x_0 = V_{0x} t$
$\Rightarrow x = 0 + (-5.78) \times 7.4$
$\therefore x = -42.77$

$y = 0$

$z - z_0 = V_{0z} t$
$\Rightarrow z = 0 + (15.89 \times 7.4)$
$\therefore z = 117.59$

$\therefore \vec{r}_B = -42.77\hat{i} + 117.59\hat{k}$

Now, displacement, $\vec{r} = \vec{r}_A - \vec{r}_B$

$= 89.024\hat{i} + 41.52\hat{k} + 42.77\hat{i} - 117.59\hat{k}$

$= 131.79\hat{i} - 76.07\hat{k}$

$|\vec{r}| = \sqrt{(131.79)^2 + (-76.07)^2}$

$= \boxed{152.17\,m}$

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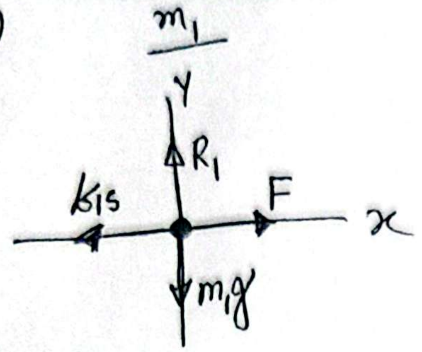
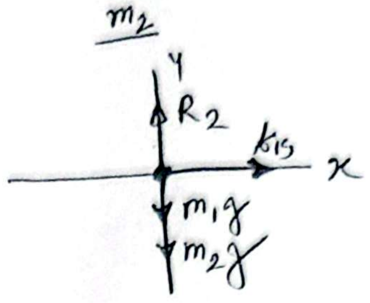
2\. (a)





(b) According to Newton's 2nd law,

For, m_1 along the y-axis, $\Sigma F_y = m_1 a_{1y}$

$\Rightarrow R_1 - m_1 g = m_1 \times 0$

$\Rightarrow R_1 = m_1 g = 2.2 \times 9.8 = 21.56$

$|\vec{a_1}| = |a_2| = a$

$a_{1x} = a,\ a_{1y} = 0$

$a_{2x} = a,\ a_{2y} = 0$

Again, along the x axis.

$\Sigma F_x = m_1 a_{1x}$

$\Rightarrow F_{max} - f_{1s} = m_1 a$

$\Rightarrow F_{max} - \mu_s R_1 = 2.2a$

$\Rightarrow F_{max} - (0.4 \times 21.56) = 2.2a$

$\Rightarrow F_{max} - 8.624 = 2.2a$ — (1)

$\Rightarrow F_{max} = (2.2 \times 1.8) + 8.624$

$\boxed{\therefore F_{max} = 12.584\,N}$

For m_2, along the x-axis

$\Sigma F_x = m_2 a_{2x}$

$\Rightarrow f_{1s} = m_2 a$

$\Rightarrow a = \frac{\mu_s R_1}{m_2}$

$\therefore a = \frac{8.624}{4.8} = \boxed{1.8\ ms^{-2}}$

(c) For m_1 along the y axis

$\Sigma F_y = m_1 a_{1y} \Rightarrow R_1 - m_1 g = m_1 \times 0 \Rightarrow R_1 = 21.56\,N$

Along the x-axis, $\Sigma F_x = m_1 a_{1x}$

$\Rightarrow 2F_{max} - f_{1K} = m_1 a_1$

$\Rightarrow a_1 = \frac{(2 \times 12.584) - (0.3 \times 21.56)}{2.2}$

$\boxed{\therefore a_1 = 8.5\ ms^{-2}}$

$\vec{a_1} = a_1\hat{i} + 0\hat{j}$

$\vec{a_2} = a_2\hat{i} + 0\hat{j}$

$f_{1K} = \mu_K R_1$

$= 0.3 \times 21.56$

$= 6.468\,N$

For m_2, $\Sigma F_x = m_2 a_{2x} \Rightarrow f_{1K} = m_2 a_2$

$\Rightarrow a_2 = \frac{6.468}{4.8} = \boxed{1.35\ ms^{-2}}$